

Hak-Bul



AI-Powered Legal Assistant for Turkish Law — RAG-based web application that answers citizens' legal questions with cited sources from Turkish legislation and Supreme Court decisions.

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Closing

Thank you for listening!

1. INTRODUCTION

1.

 1.1 PROBLEM STATEMENT

Access to legal information in Turkey is economically and practically difficult. Attorney fees are often unaffordable; hundreds of statutes, thousands of regulations, and ever-updating Supreme Court decisions make navigation nearly impossible for ordinary citizens.

-  High cost of professional legal consultation
-  Complex, fragmented Turkish legislation (33+ core laws)
-  No reliable AI tool for Turkish legal domain

 1.2 PURPOSE & TARGET AUDIENCE

Hak-Bul enables citizens to ask legal questions in plain Turkish and receive cited, source-backed answers drawn from Turkish legislation and Yargıtay (Supreme Court) decisions.

 Primary Users: Citizens with everyday legal questions (tenancy, employment, consumer rights, family law)

 Secondary Users: Small business owners, students, non-lawyers who need a structured overview of their rights.

1. INTRODUCTION

Core

 GET ANSWERS TO YOUR LEGAL QUESTIONS

Submit any legal question in plain Turkish or English and receive a cited, source-backed answer.

 COMPARE YOUR LEGAL DOCUMENTS

Upload and compare PDF legal documents side by side for differences and analysis.

 SHARE INFORMATION VIA FORUM

Post questions on the community forum and receive lawyer-verified replies.

 LEGAL DOCUMENT TEMPLATES

Generate common legal document templates via ReportLab for download.

 HakBul

Hoş geldiniz. Nasıl yardımcı olabilirim?

Hukuki sorunuzu yazın; kanun maddeleri ve Yargıtay kararlarıyla desteklenmiş bir yanıt alın.

ÖRNEK SORULAR

MEDENİ HUKUK

Boşanma davası açmak istiyorum, hangi sebeplere dayanabilirim ve süreç nasıl başlar?



MEDENİ HUKUK

Mirasçılık belgesi nasıl alınır ve miras payımı göstermek için hangi hükme bakılır?



MEDENİ HUKUK

Vasiyetnameye hangi sebeplerle itiraz edebilir veya iptal davası açabilirim?



TİCARET HUKUKU

Şirket kurmak istiyorum, kuruluş ve ticaret siciline tescil için hangi temel kurallara bakmalıyım?



TİCARET HUKUKU

Tacir sayılmanın hukuki sonuçları nelerdir, basiretli davranma ve ticaret unvanı yükümlülüğü ne demektir?



TİCARET HUKUKU

Anonim şirket genel kurul kararına karşı iptal davası açmak için kimler, hangi sürede dava açabilir?



Hukuki sorunuzu buraya yazın...

 0/1000

Enter ile gönder · Shift+Enter satır ekler

Sor ↑

Yanıtlar bilgi amaçlıdır - Avukat görüşünün yerini tutmaz

1. INTRODUCTION

1.3 Significance Of

 ACCESS TO JUSTICE

Democratizes legal knowledge by providing **free, instant** guidance to citizens who cannot afford attorneys — aligned with the UN's SDG 16 (Peace, Justice & Strong Institutions).

 AI FOR TURKISH LAW

First Turkish-specific legal RAG system with a **75,789-chunk corpus** (66,755 Supreme Court decisions + 9,034 statute articles) and verified retrieval@5 accuracy of **81.4%**.

 RESPONSIBLE AI

Integrates strict ethical guidelines and guardrails, explicitly advising users to consult professional legal counsel for high-stakes matters to ensure safety and compliance.

1. INTRODUCTION

1.4 Literature

🗨️ GENERAL PURPOSE LLMS (GPT-4)

Provide robust reasoning and natural language generation. While highly capable, they require domain-specific grounding (RAG) to ensure accuracy and provide exact citations for Turkish legal texts.

📖 DOMAIN-SPECIFIC MODELS (LEGAL-BERT)

Pioneering works by Chalkidis et al. established the effectiveness of specialized embeddings for legal NLP, primarily focusing on classification and representation in English contexts.

🏛️ TRADITIONAL LEGAL PORTALS

Official platforms like Mevzuat.gov.tr offer comprehensive, authoritative archives of Turkish law, relying on exact-match keyword search paradigms for document retrieval.

⚙️ HAK-BUL'S CONTRIBUTION

Synergizes the reasoning power of LLMs with a **custom RAG architecture**. It utilizes semantic embeddings over a vast Turkish legal corpus to deliver accurate, context-aware, and fully cited responses.

1. INTRODUCTION

1.5 Innovative

8-STEP RAG PIPELINE

Query categorization (14 legal domains) → rewriting → dual-collection Qdrant retrieval → cross-encoder reranking → Llama generation. Category penalty suppresses off-topic results by **50%**.

EXAMPLE

"İşten çıkarıldım, tazminat hakkım var mı?" → category: **İş Hukuku** → rewrite: "4857 İş Kanunu kıdem tazminatı koşulları"

REAL-TIME SSE STREAMING

Answers stream token-by-token via Server-Sent Events, providing a near-instant response feel powered by Groq's LPU inference (**Llama-3.3-70b-versatile**, 128k context window).

RELIABLE GUARDRAILS

Implements ethical AI safeguards for sensitive legal matters. Features continuous weak-query logging to monitor low-confidence retrievals and actively expand the legal corpus.

BEYOND CHAT

PDF document analysis & comparison, legal document template generation (ReportLab), community forum with lawyer-verified replies, and an admin analytics dashboard — all in one platform.

2. SOFTWARE DEVELOPMENT

2. Software

2.1 Requirements Analysis

Hak-Bul is designed as a web-based legal information assistant for Turkish law. The main requirement is to transform a user's natural-language question into a cited, understandable, non-advisory legal information response.

- Accept Turkish and English legal questions from registered users and guests.
- Retrieve relevant legislation and court-decision excerpts before generating an answer.
- Show source summaries, the legal category, confidence signals, and an informational warning.
- Support document analysis, document comparison, templates, forum activity, and admin monitoring.

2.1.1 User Requirements

Target users need fast legal orientation without learning legal terminology or manually searching many official resources.

- Ask a legal question in everyday language and receive a clear answer.
- Inspect which law article or Yargıtay decision supports the response.
- Continue conversations as a guest or authenticated user and export or share their history.
- Upload PDFs for analysis, compare two legal documents, and generate common legal drafts.

2. SOFTWARE DEVELOPMENT

2.1.2 System

Functional System Requirements

- Run the RAG pipeline: category detection, query rewriting, retrieval, optional reranking, source filtering, and answer generation.
- Provide REST and SSE endpoints for standard and streaming answers.
- Manage JWT authentication, refresh-token rotation, guest sessions, roles, and profile updates.
- Persist chat history, feedback, shared conversations, forum content, and weak retrieval queries.
- Expose admin analytics for users, categories, feedback, and low-confidence questions.

Non-Functional System Requirements

- Reliability: fall back to the local legal corpus when Qdrant is unavailable.
- Performance: stream generated tokens and keep the answer flow responsive for demos.
- Security: HttpOnly cookies for refresh/guest sessions, bcrypt password hashing, and role checks.
- Maintainability: keep routers, services, RAG modules, migrations, and frontend flow helpers separated.
- Legal safety: every answer remains informational and does not replace professional legal advice.

2. SOFTWARE DEVELOPMENT

2.2 Software

01

Iterative Process Model

The project followed an incremental approach: first a working RAG question-answering core, then authentication, history, document tools, templates, forum, admin screens, and test hardening.

MVP first, then feature expansion

02

Development Workflow

- Define the user flow and backend contract.
- Implement API/service logic and frontend pages.
- Add contract tests around critical flows.
- Document the API, database schema, RAG behavior, and operational notes.

03

Project Evidence

- The backend pytest suite covers auth, chat, retrieval, documents, feedback, forum, and admin modules.
- Frontend contract tests cover routing, authenticated sessions, chat UI, templates, profiles, forum, and admin flows.
- Docker Compose defines PostgreSQL, migration runner, backend, and frontend services.

2. SOFTWARE DEVELOPMENT

2.3 Architectural

Frontend Layer

React 19 and Vite power the single-page user interface. Pages are organized around chat, document analysis, comparison, templates, forum, profile, and admin views.

- Context providers manage language, theme, and authentication state.
- API helpers centralize request behavior and token handling.
- Flow tests protect important user journeys.

Backend and Service Layer

FastAPI exposes REST and streaming endpoints. Routers delegate business rules to services and the RAG modules, keeping responsibilities separated.

- Routers: auth, chat, documents, templates, feedback, forum, admin.
- Services: chat persistence, forum moderation, feedback, templates, documents, admin metrics.
- RAG modules: categorizer, query rewriter, retriever, reranker, generator.

Data and AI Layer

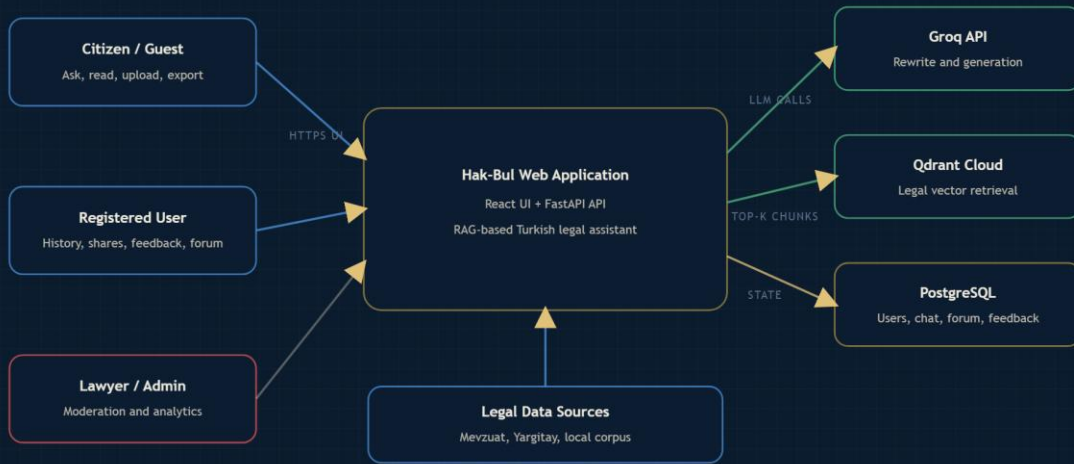
PostgreSQL stores application data, Qdrant stores legal-text embeddings, and Groq provides LLM inference for query rewriting and answer generation.

- Qdrant — dual collections: court decisions and statute articles indexed separately.
- Groq + Llama 3.3 70B: LPU architecture supports low-latency SSE streaming.
- multilingual-e5-large: strong multilingual embedding model for Turkish legal text.
- bge-reranker-v2-m3: ~270 MB cross-encoder model, toggled via RERANKER_ENABLED flag.

2. SOFTWARE DEVELOPMENT

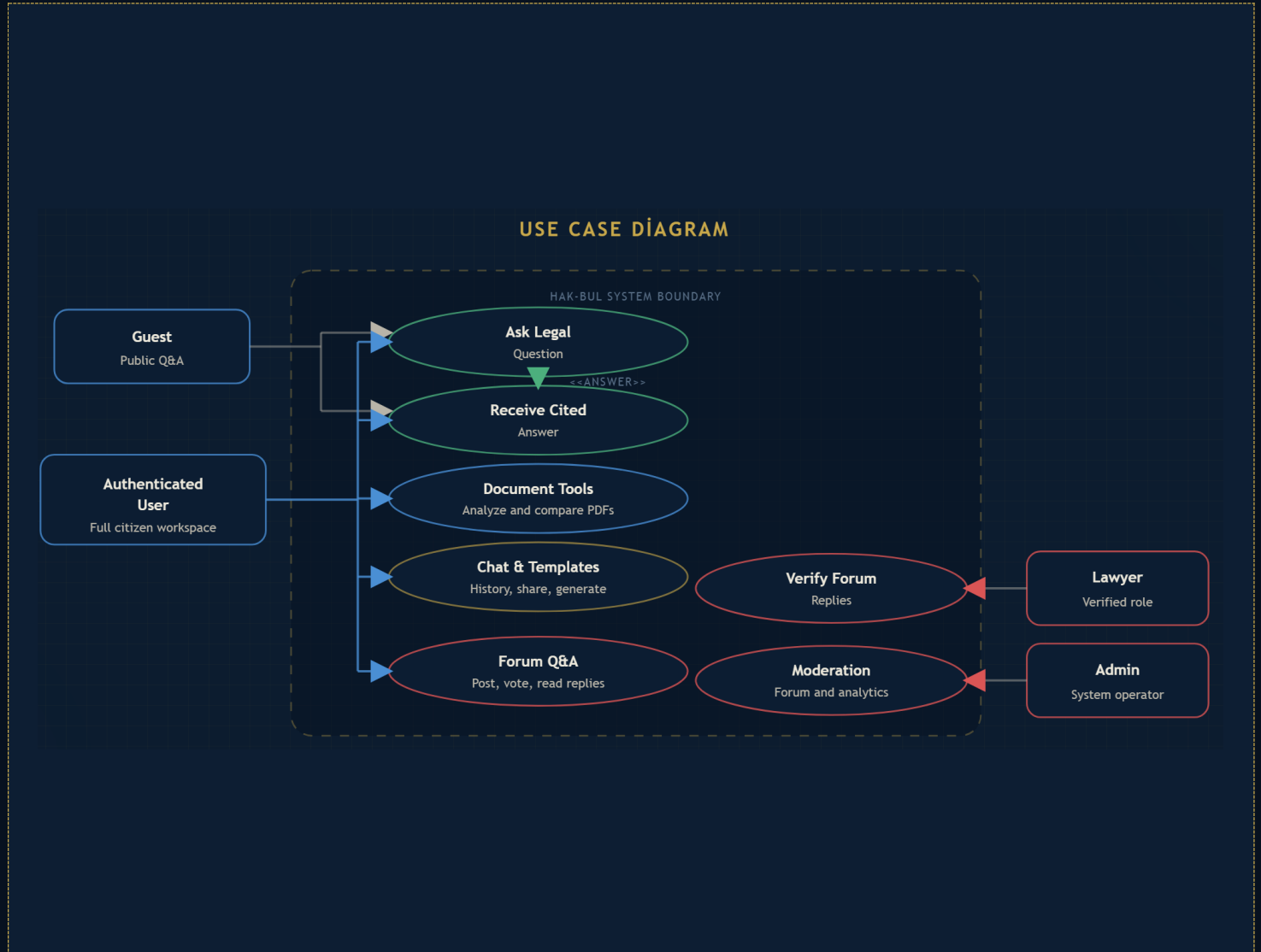
2.4 Context

SYSTEM CONTEXT



2. SOFTWARE DEVELOPMENT

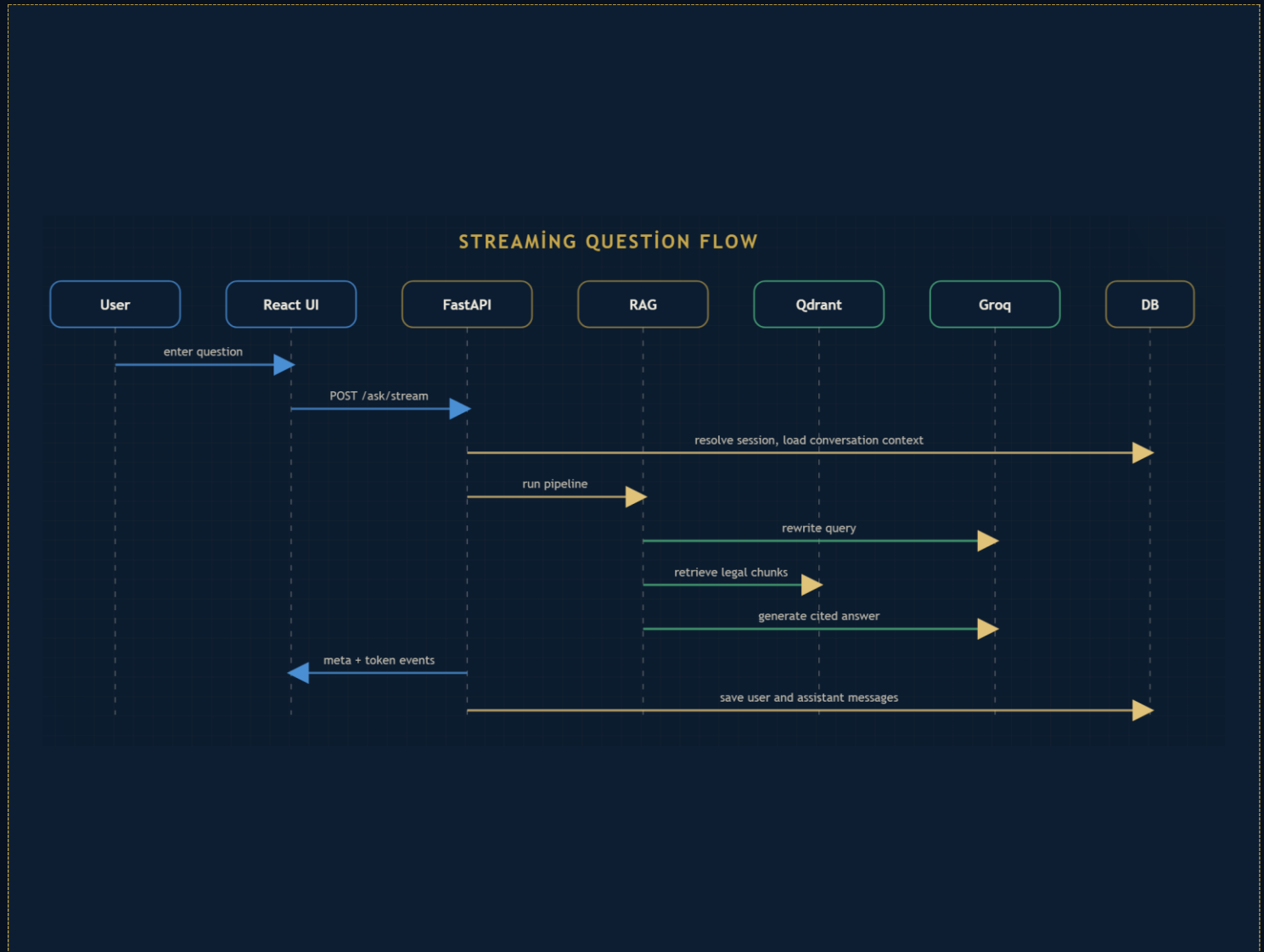
2.5 Use Case



2. SOFTWARE DEVELOPMENT

2.6 Sequence

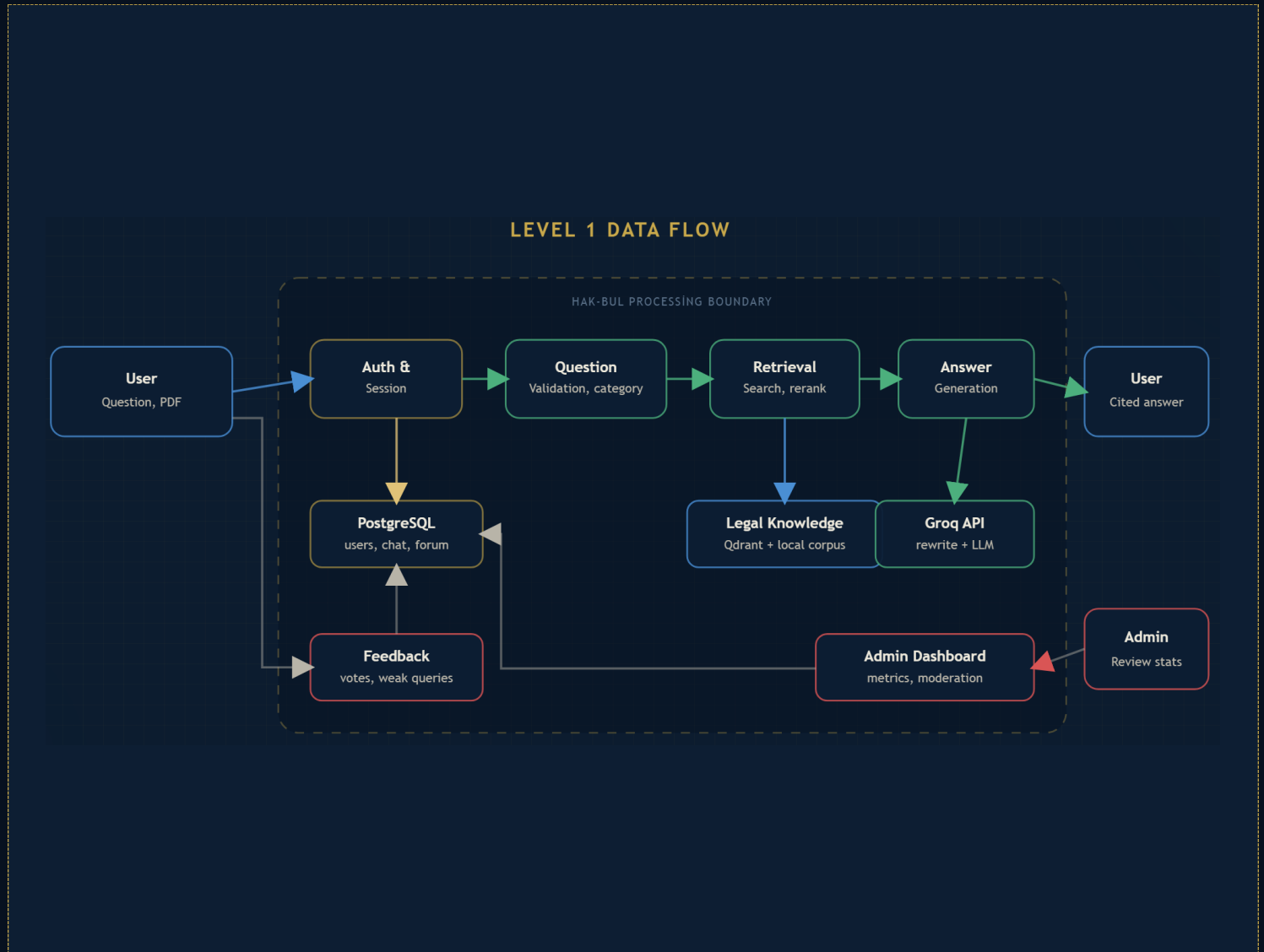
Streaming Question Flow



2. SOFTWARE DEVELOPMENT

2.7 Data Flow

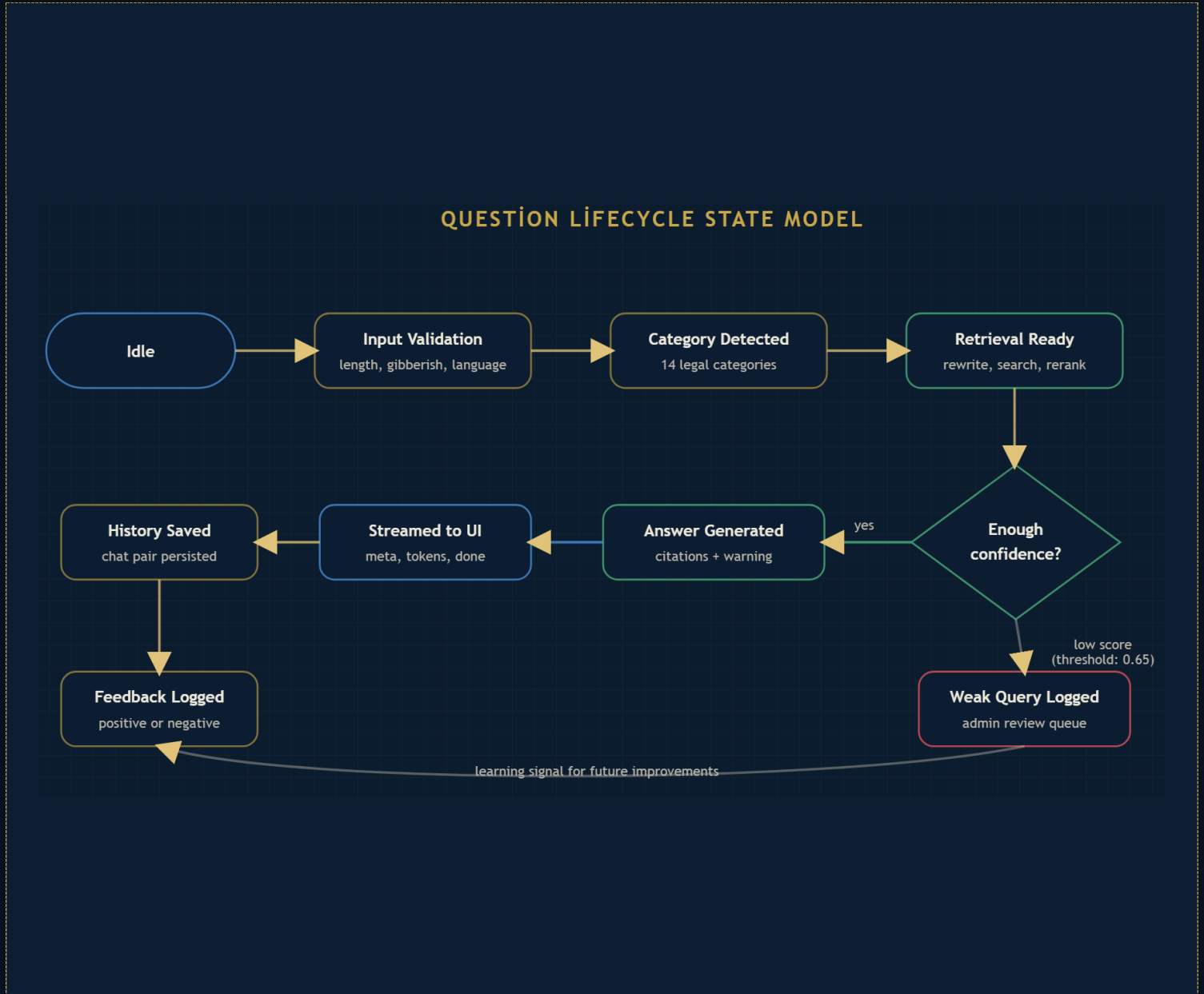
Level 1 Data Flow — Hak-Bul Processing Boundary



2. SOFTWARE DEVELOPMENT

2.8 Behavioural

Question Lifecycle State Model



3. DATABASE DESIGN

3. Database

3.1 Key Tables

PostgreSQL stores all relational application data across 9 tables managed via Alembic migrations.

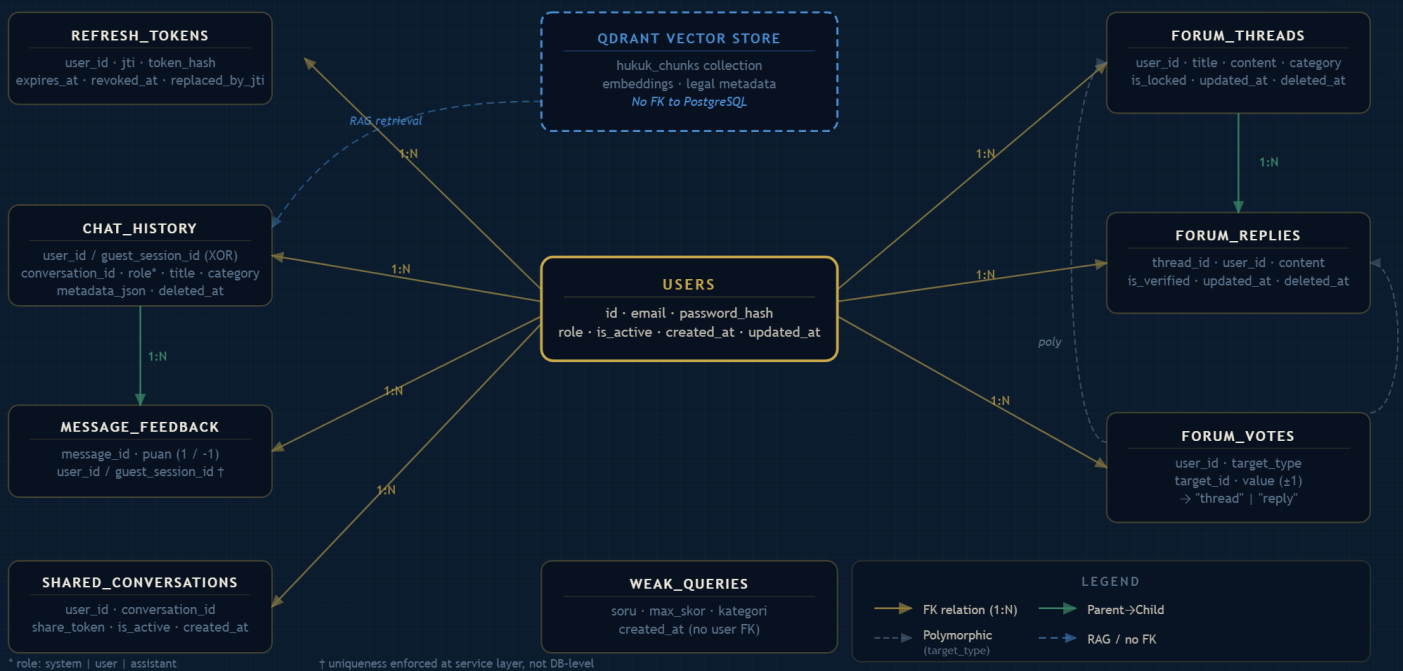
- Identity: users, refresh_tokens
- Chat: chat_history, message_feedback, shared_conversations
- Quality: weak_queries — low-confidence retrieval logging
- Forum: forum_threads, forum_replies, forum_votes

Database Notes

- 10 Alembic migrations (0001–0010) govern the schema.
- Soft-delete via deleted_at is used in chat_history, forum_threads, and forum_replies.
- chat_history and message_feedback support both authenticated users and guests (XOR on user_id / guest_session_id).
- message_feedback uniqueness is enforced at the service layer, not DB-level constraint.
- Qdrant is separate: vector store for legal chunks — no FK to PostgreSQL.

3. DATABASE DESIGN

3.2 ER



4. RESULT

4.1 Used Technologies

 PROGRAMMING LANGUAGES Python 3.11 powers the backend and RAG pipeline. JavaScript/JSX builds the frontend. SQL is used for relational persistence and migrations.	 FRAMEWORKS FastAPI, SQLAlchemy, and Alembic form the API and database layer. React 19, Vite, Tailwind CSS, and ESLint support the user interface.
 DATABASES / RETRIEVAL PostgreSQL stores application records. Qdrant performs vector search over legal chunks. Local JSON corpus fallback supports development and resilience.	 SECURITY / AI JWT access tokens, refresh rotation, HttpOnly cookies, and bcrypt protect accounts. Groq Llama-3.3-70b generates answers; multilingual-e5-large handles embeddings; bge-reranker-v2-m3 reranks results.

Retrieval Evaluation — 167 Test Questions · 14 Legal Categories

81.4% retrieval@5 (law)	52.9% retrieval@5 (article)	67% → 81.4% Phase 1 → Phase 2	85%+ target (corpus complete)
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-  **Best:** Civil, Commercial, Social Security — 100% law accuracy
-  **Mid:** Criminal, Consumer, IT Law — 83–92% law accuracy
-  **Gap:** Constitutional — 33.3% (missing statutes in corpus)

4. RESULT

4.2 Conclusion and

✓ CONCLUSION

Hak-Bul delivers a source-backed legal assistant for Turkish law with chat, streaming answers, PDF analysis, document templates, forum workflows, and admin analytics.

✓ 75,789 legal chunks indexed

✓ retrieval@5 accuracy: **81.4%**

✓ 8-step RAG pipeline — demo ready

🔍 FUTURE PLAN 1 — RETRIEVAL QUALITY

- Add 18 missing statutes (KVKK, FSEK, 5651, Constitutional Court laws, VUK) to reach 85%+ retrieval accuracy.
- Optimize chunk size for VUK (tax law) — current granularity limits article-level retrieval.
- Evaluate hybrid search: BM25 + vector for improved keyword-sensitive queries.
- Leverage weak-query logs as a feedback loop for targeted corpus expansion.

🚀 FUTURE PLAN 2 — PRODUCTION HARDENING

- Add monitoring, retry logic, and failover behavior for Qdrant and Groq API outages.
- Implement response caching for high-frequency legal queries.
- Deployment smoke tests and operational dashboards for uptime visibility.
- Enable reranker by default in production (RERANKER_ENABLED=true) once latency benchmarks confirm acceptable response times.

Thank you for listening!

Hak-Bul

Contact / team / institution information placeholder.